

3.5 Newton's laws of motion and momentum

This section provides knowledge and understanding of Newton's laws – fundamental laws that can be used to predict the motion of all colliding or interacting objects in applications such as sport (HSW1, 2). Newton's law can also be used to understand some of the safety features in cars, such as air bags, and to evaluate the benefits and risks of such features (HSW9). Learners should be aware that the introduction of mandatory

safety features in cars is a consequence of the scientific community analysing the forces involved in collisions and investigating potential solutions to reduce the likelihood of personal injury (HSW10, 11, 12).

There are many opportunities for learners to carry out experimental work and analyse data using ICT techniques (HSW3).

3.5.1 Newton's laws of motion

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) Newton's three laws of motion	HSW7
(b) linear momentum; $p = mv$; vector nature of momentum	
(c) net force = rate of change of momentum; $F = \frac{\Delta p}{\Delta t}$	Learners are expected to know that $F = ma$ is a special case of this equation. HSW9, 10 M2.1, M3.9
(d) impulse of a force; impulse = $F\Delta t$	
(e) impulse is equal to the area under a force–time graph.	Learners will also be expected to estimate the area under non-linear graphs. HSW3 Using a spreadsheet to determine impulse from F – t graph. M3.8, M4.3